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MANUFACTURE AND TEST SEQUENCE

IN ORDER

1	MACHINE CP-101 body and CP-102 cover. Pocket, plenums, ports, mount holes.
2	SKIVE FINS 125 fins, 0.15 thk x 4.0 tall @ 0.40 pitch. one pass per fin
3	DEBURR + CLEAN Ultrasonic in alkaline, DI rinse, dry. Borescope every channel. no cutting fluid
4	BRAZE Vacuum or dry N2/H2. BCuP-5 preform on the perimeter land ONLY. fin tips NOT brazed
5	PROOF PRESSURE Hydrostatic 3 x working, 5 min, no permanent set. typically 15 bar
6	LEAK TEST Helium mass spectrometer under vacuum. < 1e-8 Pa.m3/s
7	FLOW TEST Measure dP at 1.5 L/min. Record as the commissioning baseline. 6.1 kPa nom
8	THERMAL TEST Die simulator at 700 W. Verify R <= 0.032 K/W. first article + AQL
9	FLUSH + DRY DI flush, N2 purge, cap both ports. ship dry, capped
10	INSTALL Mount per CP-200. Fill, vent, run 30 min, re-vent. air blankets fins

ACCEPTANCE		
TEST	LIMIT	APPLIED TO
Helium leak	< 1e-8 Pa.m3/s	100 %
Proof pressure	3 x working, no set	100 %
dP at 1.5 L/min	6.1 kPa +20 % / -30 %	100 %
Thermal R, die simulator	<= 0.032 K/W at 700 W	first article + AQL

COOLANT SPECIFICATION		
PROPERTY	SPECIFICATION	WHY
Fluid	PG25, 25 % propylene glycol	freeze + biocide
Inhibitor package	phosphate/azole, Cu-rated	protects copper
pH	8.0 - 9.5	copper passivity
Conductivity	< 50 uS/cm	galvanic / leak sensing
Particulate	filtered 50 um at the CDU	0.25 mm channels
Supply temperature	45 C nominal, 50 C max	warm-water loop
Flow per plate	1.5 L/min	design point
Change interval	per fluid supplier, test annually	inhibitor depletes

WETTED MATERIALS — GALVANIC COMPATIBILITY		
MATERIAL	USE	VERDICT
Copper C10200, brass, bronze	plate, fittings	ALLOWED
Stainless 304 / 316	tubing, QD bodies	ALLOWED
EPDM	hose, external seals	ALLOWED
ALUMINIUM, any alloy	anywhere wetted	PROHIBITED
Zinc, galvanised steel	anywhere wetted	PROHIBITED
Nitrile, PTFE tape	seals	not recommended

Copper is strongly cathodic. One aluminium fitting in a copper loop corrodes

through in months and the debris then blocks the 0.25 mm channels.

COMMISSIONING

- Fill slowly from the lowest point. Vent at the highest point until clear.
- Run 30 minutes, re-vent, then record dP at 1.5 L/min as the baseline for this plate.
- Trend dP, not only temperature. Fouling shows in dP long before the die gets hot.
- Service trigger: dP 20 % above the commissioning baseline at the same flow.

INDIA-VC THERMAL HARDWARE				ISO 128 / ISO 2768-mK	
TITLE PROCESS, TEST & COOLANT SPECIFICATION CP-100 COLD PLATE / MANDATORY SEQUENCE AND LOOP CHEMISTRY			SCALE NONE THIRD ANGLE PROJECTION	UNITS mm	
DRAWING No. CP-300			REV A	SHEET 6 / 6	
DRAWN AI-ASSISTED DRAFT	CHECKED — PENDING —	MATERIAL SEE BOM	FINISH SEE NOTES	DATE 2026-09-10	